MATH 152 **QUIZ 1A** NAME Answer Key

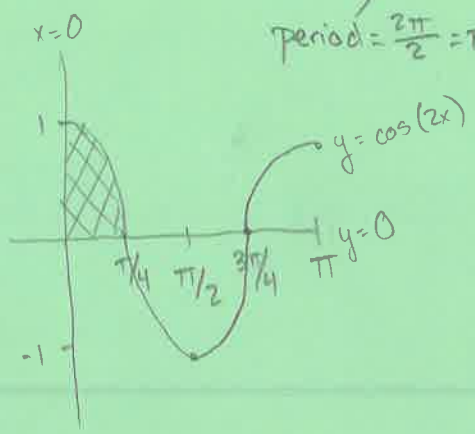
Mrs. Bonny Tighe 6.1-6.2

25 points TA_____ Wed 9/6/2017

1. Find the area of the region enclosed by the given curves. Sketch the graphs.

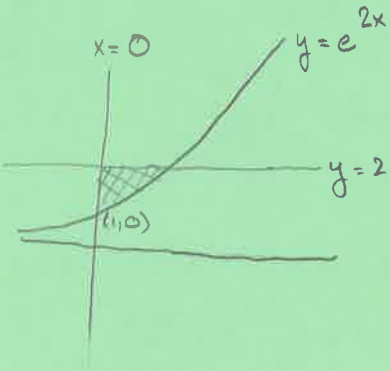
a) $y = \cos 2x$,, $y = 0$, *and* $x = 0$

period $= \frac{2\pi}{2} = \pi$



$$A = \int_0^{\pi/4} \cos(2x)\,dx$$
$$= \left(\tfrac{1}{2}\sin(2x)\right|_0^{\pi/4}$$
$$= \tfrac{1}{2}\sin\left(2\cdot\tfrac{\pi}{4}\right) - \tfrac{1}{2}\sin(2\cdot 0)$$
$$= \tfrac{1}{2}\sin(\pi/2) - \tfrac{1}{2}\sin(0)$$
$$= \tfrac{1}{2}(1) - \tfrac{1}{2}(0)$$
$$= \boxed{\tfrac{1}{2}}$$

b) $y = e^{2x}$, $y = 2$, *and* $x = 0$



Points of intersection

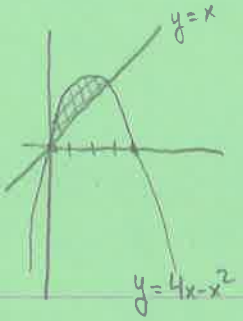
$e^{2x} = 2$

$2x = \ln 2$

$x = \frac{1}{2}\ln 2$

$$A = \int_0^{\frac{1}{2}\ln 2} (\text{top} - \text{bottom})\,dx$$
$$= \int_0^{\frac{1}{2}\ln 2} (2 - e^{2x})\,dx$$
$$= \left(2x - \tfrac{1}{2}e^{2x}\right|_0^{\frac{1}{2}\ln 2}$$
$$= 2\left(\tfrac{1}{2}\ln 2\right) - \tfrac{1}{2}e^{2(\frac{1}{2}\ln 2)} - 2(0) + \tfrac{1}{2}e^{2(0)}$$
$$= \ln 2 - \tfrac{1}{2}e^{\ln 2} - 0 + \tfrac{1}{2}e^{0}$$
$$= \ln 2 - \tfrac{1}{2}(2) + \tfrac{1}{2}$$
$$= \ln 2 - 1 + \tfrac{1}{2}$$
$$= \boxed{\ln 2 - \tfrac{1}{2}}$$

c) $y = x, \quad y = 4x - x^2$



$$A = \int_0^3 (\text{top} - \text{bottom})\,dx$$
$$= \int_0^3 (4x - x^2 - x)\,dx$$
$$= \int_0^3 (3x - x^2)\,dx$$
$$= \left(\frac{3}{2}x^2 - \frac{1}{3}x^3\right)\Big|_0^3$$
$$= \frac{3}{2}(3)^2 - \frac{1}{\cancel{3}}(3)^{\cancel{3}\,2} - 0 + 0$$
$$= 3^2\left(\frac{3}{2} - 1\right)$$
$$= 9\left(\frac{1}{2}\right) = \boxed{\frac{9}{2}}$$

Points of intersection:

$x = 4x - x^2$

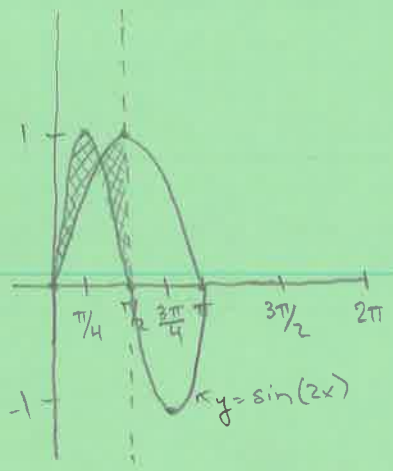
$0 = 3x - x^2$

$0 = x(3 - x)$

$x = 0, 3$

d) $y = \sin x$ and $y = \sin 2x$, $0 \le x \le \pi/2$

period $= \frac{2\pi}{2} = \pi$



$$A = \int_0^{\pi/3} (\text{top} - \text{bottom})\,dx + \int_{\pi/3}^{\pi/2} (\text{top} - \text{bottom})\,dx$$
$$= \int_0^{\pi/3} (\sin(2x) - \sin x)\,dx + \int_{\pi/3}^{\pi/2} (\sin x - \sin(2x))\,dx$$
$$= \left[-\frac{1}{2}\cos(2x) + \cos x\right|_0^{\pi/3} + \left[-\cos x + \frac{1}{2}\cos(2x)\right|_{\pi/3}^{\pi/2}$$
$$= -\frac{1}{2}\cos\left(\frac{2\pi}{3}\right) + \cos\left(\frac{\pi}{3}\right) + \frac{1}{2}\cos(0) - \cos(0) - \cos(\pi/2)$$
$$+ \frac{1}{2}\cos(\pi) + \cos(\pi/3) - \frac{1}{2}\cos(2\pi/3)$$
$$= -\frac{1}{2}\left(-\frac{1}{2}\right) + \frac{1}{2} + \frac{1}{2} - 1 - 0 + \frac{1}{2}(-1) + \frac{1}{2} - \frac{1}{2}\left(-\frac{1}{2}\right)$$
$$= \frac{1}{4} + \frac{1}{4}$$
$$= \boxed{\frac{1}{2}}$$

Point of intersection:

$\sin x = \sin(2x)$

$\sin x = 2\sin x \cos x$

$0 = 2\sin x \cos x - \sin x$

$0 = \sin x\,[2\cos x - 1]$

$\sin x = 0$

$x = 0, \cancel{\pi}, \cancel{2\pi}, \ldots$

crossed out since out of bounds

$2\cos x - 1 = 0$

$\cos x = \frac{1}{2}$

$x = \pi/3, \cancel{\frac{5\pi}{3}}$

MATH 152 **QUIZ 1** NAME Answer Key

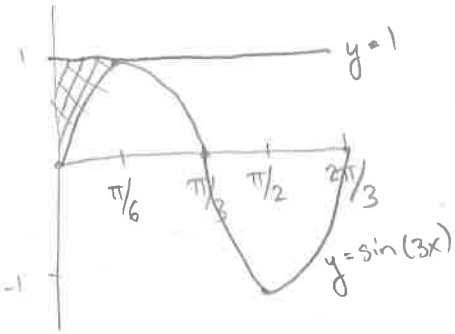
Mrs. Bonny Tighe 6.1

25 points TA_____ Wed 9/6/2017

1. Find the area of the region enclosed by the given curves. Sketch the graphs.

a) $y = \sin 3x, \quad y = 1$ *and* $x = 0$

period $= \frac{2\pi}{3}$



Point of intersection

$\sin(3x) = 1$

$3x = \pi/2$

$x = \pi/6$

$$A = \int_0^{\pi/6} (\text{top} - \text{bottom})\,dx$$

$$= \int_0^{\pi/6} (1 - \sin(3x))\,dx$$

$$= \left(x + \frac{1}{3}\cos(3x)\right)\Big|_0^{\pi/6}$$

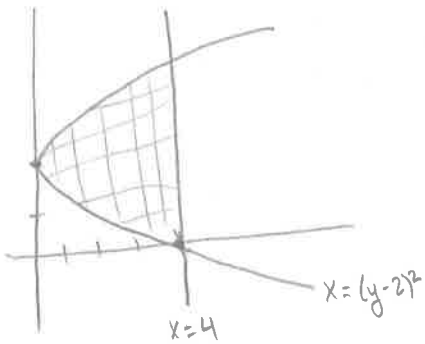
$$= \frac{\pi}{6} + \frac{1}{3}\cos\left(3\cdot\frac{\pi}{6}\right) - 0 - \frac{1}{3}\cos(0)$$

$$= \frac{\pi}{6} + \frac{1}{3}\cos(\pi/2) - \frac{1}{3}(1)$$

$$= \frac{\pi}{6} + \frac{1}{3}(0) - \frac{1}{3}$$

$$= \boxed{\frac{\pi}{6} - \frac{1}{3}}$$

b) $x = (y-2)^2$ *and* $x = 4$



Points of intersection:

$(y-2)^2 = 4$

$y - 2 = \pm 2$

$y - 2 = 2$ and $y - 2 = -2$

$y = 4$ $\quad y = 0$

$$A = \int_0^4 (\text{right} - \text{left})\,dy$$

$$= \int_0^4 \left(4 - (y-2)^2\right)dy$$

$$= \left(4y - \frac{1}{3}(y-2)^3\right)\Big|_0^4$$

$$= 4(4) - \frac{1}{3}(4-2)^3 - 0 + \frac{1}{3}(0-2)^3$$

$$= 16 - \frac{1}{3}(2)^3 + \frac{1}{3}(-2)^3$$

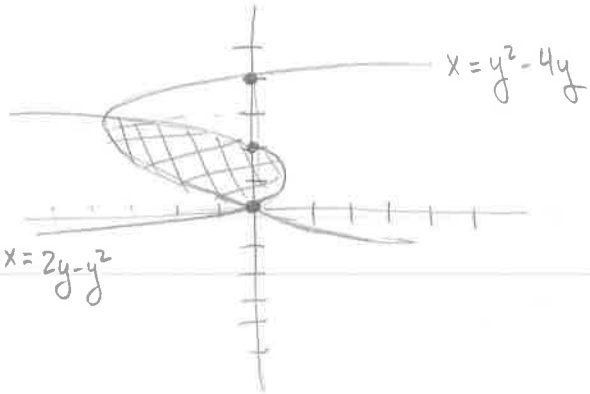
$$= 16 - \frac{1}{3}[8 + 8]$$

$$= 16 - \frac{16}{3}$$

$$= 16\left(1 - \frac{1}{3}\right)$$

$$= 16\left(\frac{2}{3}\right) = \boxed{\frac{32}{3}}$$

c) $y^2 - 4y = x$ and $2y - y^2 = x$

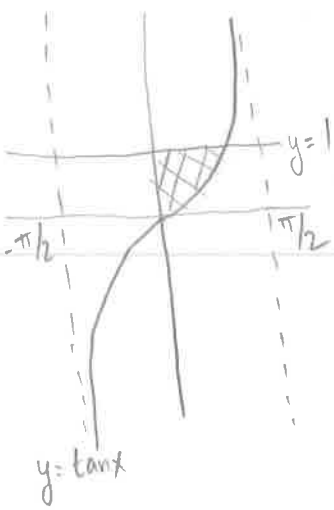


$$A = \int_0^3 (\text{right} - \text{left})\, dy$$
$$= \int_0^3 (2y - y^2 - y^2 + 4y)\, dy$$
$$= \int_0^3 (6y - 2y^2)\, dy$$
$$= \left(3y^2 - \frac{2}{3}y^3\right)\Big|_0^3$$
$$= 3(3)^2 - \frac{2}{\cancel{3}}(3)^{\cancel{3}\,2} - 0$$
$$= 3^2(3-2)$$
$$= \boxed{9}$$

Points of intersection

$$y^2 - 4y = 2y - y^2$$
$$2y^2 - 6y = 0$$
$$2y(y-3) = 0$$
$$y = 0, 3$$

d) $y = \tan x$, $x = 0$ and $y = 1$



$$A = \int_0^{\pi/4} (\text{top} - \text{bottom})\, dx$$
$$= \int_0^{\pi/4} (1 - \tan x)\, dx$$
$$= \left(x - \ln|\sec x|\right)\Big|_0^{\pi/4}$$
$$= \frac{\pi}{4} - \ln|\sec(\pi/4)| - 0 + \ln|\sec 0|$$
$$= \frac{\pi}{4} - \ln|\sqrt{2}| + \ln|1|$$
$$= \boxed{\frac{\pi}{4} - \ln(\sqrt{2})}$$

Aside:

$$\int \tan x\, dx = \int \frac{\sin x}{\cos x}\, dx \qquad u = \cos x, \quad du = -\sin x\, dx$$
$$= -\int \frac{du}{u}$$
$$= -\ln|u| + c$$
$$= -\ln|\cos x| + c$$
$$= \ln|\sec x| + c$$

Point of intersection

$$\tan x = 1$$
$$x = \pi/4, \cancel{5\pi/4}$$

not in our bound from looking at graph